IGCSE 0607 Practice Questions — E1.1

Types of number

Total Questions: 21 — **Total Marks:** 130 Marks

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1 **(a)** Express 504 as a product of its prime factors.

..... **[2]**

(b) Find the smallest positive integer k such that $504k$ is a square number.

$k =$ ..... **[2]**

2 **(a)** Write 84 as a product of its prime factors.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 126.

..... **[2]**

3 Here is a list of numbers.

2 9 12 23 36 45 64 81

(a) Write down a number from the list which is:

(i) a multiple of 12,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) both a prime number and a factor of 46.

..... **[1]**

(b) Find the lowest common multiple (LCM) of 36 and 64.

..... **[2]**

4 Here is a list of numbers.

$$0.25 \quad \sqrt{7} \quad 8 \quad 9 \quad 13 \quad -5 \quad 21 \quad 64$$

(a) From the list, write down a prime number.

..... **[1]**

(b) From the list, write down an irrational number.

..... **[1]**

(c) From the list, write down an integer less than zero.

..... **[1]**

(d) From the list, write down the reciprocal of 4.

..... **[1]**

(e) From the list, write down a number that is both a square number and a cube number.

..... **[1]**

5 Here is a list of numbers.

8 15 23 27 36 49 64

(a) From the list, write down

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) a number that is both a square number and a cube number.

..... **[1]**

(b) Express 36 as a product of its prime factors.

..... **[2]**

6 Here is a list of numbers.

2 9 15 16 23 27 64 72

(a) From the list, write down

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) a number that is both a square number and a cube number,

..... **[1]**

(iv) a triangle number.

..... **[1]**

(b) Express 72 as a product of its prime factors.

..... **[2]**

7 **(a)** Write 168 as a product of its prime factors.

..... **[2]**

(b) Find the highest common factor (HCF) of 168 and 252.

..... **[2]**

(c) Find the lowest common multiple (LCM) of 168 and 252.

..... **[2]**

8 Here is a list of numbers.

$$-5 \qquad -1 \qquad \frac{3}{8} \qquad 8 \qquad 13 \qquad 25 \qquad \sqrt{30} \qquad \pi$$

(a) From this list, write down:

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) an irrational number,

..... **[1]**

(iv) an integer that is not a natural number.

..... **[1]**

(b) Find the reciprocal of 0.8.
Give your answer as a fraction in its simplest form.

..... **[2]**

9 Two numbers, A and B, are written as products of their prime factors.

$A = 2^3 \times 3^2 \times 5$

$B = 2^2 \times 3^4 \times 7$

(a) Find the highest common factor (HCF) of A and B.

..... **[2]**

(b) Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of prime factors.

..... **[2]**

(c) The integer C is such that:

- the HCF of A and C is 6
- the LCM of A and C is $2^3 \times 3^2 \times 5 \times 11$

Find the value of C.

$C =$ ..... **[3]**

10 **(a)** Let x be a rational number and y be an irrational number.
State whether $x + y$ is rational or irrational.

..... **[1]**

(b) Give an example of an irrational number y to show that y^2 can be a rational number.

[1]

(c) The nth term of a sequence is given by the expression

$n^2 + n + 41$

(i) Work out the 3rd term of this sequence, and state whether it is a prime number.

..... **[2]**

(ii) Explain why the 41st term of this sequence is not a prime number.

..... **[2]**

11 The number N is written as a product of its prime factors as $N = 2^3 \times 3^x \times 5^2$.

(a) The highest common factor (HCF) of N and 540 is 180.
Find the value of x.

$x =$ ..... **[3]**

(b) The lowest common multiple (LCM) of N and a number M is $2^4 \times 3^3 \times 5^2 \times 7$.
Given that $M = 2^y \times 3^z \times 7$, find the value of y and the value of z.

$y =$ ..... , $z =$ ..... **[2]**

(c) Find the smallest positive integer k such that Nk is a cube number.

$k =$ ..... **[2]**

12 **(a)** The highest common factor (HCF) of x and 12 is 4.
The lowest common multiple (LCM) of x and 12 is 120.
Find the value of x.

$x =$ ..... **[2]**

(b) **(i)** Express 1008 as a product of its prime factors.

..... **[2]**

(ii) An integer N is a perfect cube.
The highest common factor (HCF) of N and 1008 is 72.
Find the smallest possible positive value of N.

$N =$ ..... **[3]**

13 Let $m = p^3q$ and $n = pq^3$, where p and q are distinct prime numbers.

(a) Express, in terms of p and q:

(i) the highest common factor (HCF) of m and n,

..... **[1]**

(ii) the lowest common multiple (LCM) of m and n.

..... **[1]**

(b) The HCF of m and n is 15.
It is given that $m > n$.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(c) Show algebraically that the product $m \times n$ is a square number.

[2]

(d) Using your values of p and q, find the smallest positive integer k such that $m \times k$ is a cube number.

$k =$ ..... **[2]**

14 **(a)** The lowest common multiple (LCM) of two numbers p and q is 1260.
The highest common factor (HCF) of p and q is 14.
Given that $p = 70$, find the value of q.

$q =$ ..... **[3]**

(b) Consider the two numbers M and N given by:

$M = 2^x \times 3^2 \times 5$

$N = 2^3 \times 3^y \times 7$

The highest common factor (HCF) of M and N is 72.
The lowest common multiple (LCM) of M and N is 10080.
Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) Given that n is a positive integer, explain why $n^2 + n$ is always an even number.

.....

.....

..... **[2]**

15 Three numbers A, B and C are given as products of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^x \times 5^2 \qquad C = 2^y \times 3 \times 7$$

The highest common factor (HCF) of A and B is 180.
The lowest common multiple (LCM) of B and C is 12600.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

16 Two numbers, A and B, are given as a product of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^3 \times 7$$

(a) Find the highest common factor (HCF) of A and B.

..... **[1]**

(b) Find the lowest common multiple (LCM) of A and B.
Give your answer as a product of its prime factors.

..... **[2]**

(c) $A \times k$ is a perfect square.
Find the smallest positive integer value of k.

$k =$ ..... **[2]**

17 Number A and Number B are given as products of their prime factors.

$A = 2^2 \times 3^3 \times 5$

$B = 2^x \times 3 \times 7$

(a) The highest common factor (HCF) of A and B is 6.
Find the value of x.

$x =$ ..... **[2]**

(b) Number $C = 2^2 \times 3^y \times 5^2$.
The lowest common multiple (LCM) of A and C is 8100.
Find the value of y.

$y =$ ..... **[3]**

(c) Here is a list of numbers.

$$\sqrt{11} \qquad -3.5 \qquad 4\pi \qquad \frac{7}{9} \qquad \sqrt{144} \qquad 0.\dot{4}$$

Write down all the irrational numbers from this list.

..... **[2]**

18 **(a)** **(i)** Write down a rational number between 0.4 and 0.5.

..... **[1]**

(ii) Write down an irrational number between 4 and 5.

..... **[1]**

(b) Give an example to show that the sum of two different prime numbers can be a square number.

[2]

(c) Write down the 6th triangle number.

..... **[1]**

(d) Find the highest common factor (HCF) of $24a^3b^2$ and $36a^2b^4$.

..... **[2]**

19 **(a)** Find the highest common factor (HCF) of 84 and 120.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 120.

..... **[2]**

(c) Write down the reciprocal of 0.8 as a fraction in its simplest form.

..... **[2]**

(d) State whether $\sqrt{120}$ is rational or irrational.
Give a reason for your answer.

..... **[2]**

20 **(a)** x, y and z are positive integers.

- x is a prime number.
- y is a cube number greater than 1.
- z is a triangle number.

It is given that $x + y = z$ and $x < 10$.

Find two possible sets of values for x, y and z.

(i) First set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

(ii) Second set of values.

$x =$ ..... , $y =$ ..... , $z =$ ..... **[2]**

(b) p and q are two different irrational numbers.

Write down an example of p and q such that:

(i) $p + q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

(ii) $p \times q$ is a rational number.

$p =$ ..... , $q =$ ..... **[1]**

21 **(a)** The number 168 can be written as $2^3 \times 3 \times 7$.
The number N can be written as $2^x \times 3^2 \times 7^y$, where x and y are integers.
The highest common factor (HCF) of 168 and N is 84.
The lowest common multiple (LCM) of 168 and N is 3528.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) Find the exact value of N.

$N =$ ..... **[1]**

(c) Two flashing lights, A and B, are turned on at the same time.
Light A flashes every 45 seconds.
Light B flashes every 72 seconds.
They both flash together at 10:15:00.
Work out the next time they flash together.

..... **[3]**